

EMIRCAN TUTAR

Software Engineer

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emirtutar.github.io/Portfolio/

SUMMARY

Reliable, detail-oriented Software Engineer with around one year of professional experience in the automotive sector (Continental), focused on backend development, test automation, and CI/CD. Strong in Java, Python, and C++, with hands-on experience designing Docker-based, CI-ready test infrastructure and integrating automated pipelines with GitHub Actions, Jenkins, and Azure DevOps. Comfortable across the stack, REST APIs, relational and NoSQL databases, and containerized distributed systems and used to delivering well-tested results in agile Scrum teams. Backed by a portfolio of self-built projects spanning mobile, web, computer vision, and robotics. All projects: emirtutar.github.io/Portfolio/#/projects

SKILLS

Programming Languages:	Advanced: Java, Python, C++, SQL Familiar: C#, JavaScript, HTML/CSS
Backend & Frameworks:	Spring, Java EE, REST APIs, Maven, Junit, CMake, eCAL, ROS
DevOps & CI/CD:	Git, Docker, Kubernetes, Jenkins, GitHub Actions, Azure DevOps, CI/CD
Databases:	SQL, Oracle, Firebase / Firestore (NoSQL)
Testing & Quality:	Unit, Integration, System, UI Testing, Robot Framework, pytest, gtest
Frontend:	Angular, React, Vue.js, Node.js
Tools & Methods:	Jira, Agile / Scrum
AI-Assisted Development:	Claude Code, ChatGPT, Gemini
Languages:	German (native), English (fluent), Turkish (native)

EXPERIENCE

Software Test Engineer (incl. Bachelor's Thesis)

Continental – ADC Automotive Distance Control Systems, Lindau · 03/2025 – 09/2025

Bachelor's Thesis: "Developing a Concept for Automated Testing of IPC Middleware – Demonstrated with eCAL"

- Built a fully automated test infrastructure for IPC/middleware that cut manual testing time by roughly 70% and made regression tests reproducible.
- Developed Docker- and Robot-Framework-based system and integration tests — isolated, parallelizable, and CI-ready.
- Designed around 40 test cases covering publish/subscribe, RPC, and crash and network-failure scenarios to safeguard critical communication paths.
- Integrated test execution into CI/CD pipelines (GitHub Actions), catching errors at commit time instead of late in the release cycle.

More details: emirtutar.github.io/Portfolio/#/projects/ecal-test-suite

Technologies: Agile/Scrum, Java, Python, C++, Docker, Robot Framework, Git, GitHub Actions, CI/CD, eCAL, IPC / middleware, distributed systems, TCP/UDP, shared memory, test automation, test driven development

Software Engineer – Software Bring-Up & Test (Internship Semester)

Continental – ADC Automotive Distance Control Systems, Lindau · 09/2024 – 03/2025

- Integrated software and hardware components for sensor data processing and inter-process communication (IPC).
- Developed and optimized C++ and Python modules, improving the runtime and maintainability of the processing pipeline.
- Applied eCAL middleware (publisher–subscriber model) and validated software in simulation and on real test drives.
- Automated build and test processes with Jenkins (CI/CD), largely eliminating manual builds.

Technologies: Jira, Agile/Scrum, C++, Python, Jenkins, CI/CD, eCAL, Git, system integration, sensor data processing, 3D simulation, middleware

Java Programming Tutor

RWU – Ravensburg-Weingarten University: 09/2022 – 02/2023

- Supported students with practical Java lab exercises, debugging, and troubleshooting.
- Taught core programming and object-oriented design concepts and helped students structure their solutions.

Technologies: Java, Spring, Maven, Object-Oriented Programming, Debugging

Production and Assembly Assistant

ZF Friedrichshafen, Friedrichshafen: 09/2021 – 10/2021, 08/2022 – 09/2022, 08/2023 – 10/2023, 05/2026 – 08/2026

- Quality assurance of planetary carriers in series production; operated and maintained KUKA robots and press machines.

PROJECTS

RateMe - Android Product Rating App (Team project)

Barcode-scanning Android app with product lookup and a Firebase-backed rating system; email-verified authentication, a real-time Firestore backend, favourites, scan history, and a multi-user comment feed.

- **More details:** emirtutar.github.io/Portfolio/#/projects/rateme
- **Technologies:** Android, Java, Firebase Auth, Firestore (NoSQL), REST API, Backend, Frontend, Git

Patient Monitoring Prototype (Team project)

Camera-based fall and bed-exit detection using YOLOv5 on Raspberry Pi; MQTT messaging layer with encrypted staff alerts, packaged as a full Docker-Compose stack with hardware alarms (LEDs, buzzer).

- **More details:** emirtutar.github.io/Portfolio/#/projects/patient-monitoring
- **Technologies:** Python, Docker, Raspberry Pi, YOLOv5, MQTT, Git

AI-Assisted Software Testing

Evaluated GPT4o for automated test generation across unit, integration, and UI tests on three Java projects; measured code coverage, error rates, and prompt effort, documenting strengths and limits.

- **More details:** emirtutar.github.io/Portfolio/#/projects/java-testing
- **Technologies:** Java, Maven, Spring, Junit, JavaFX, TestFX, GPT4o, Git

Outdoor Planner – Weather-Aware Scheduler (Team project)

Vue.js single-page app combining live weather data (OpenWeatherMap) with an outdoor appointment manager; responsive layout and full appointment CRUD. Live at outdoorplanner.netlify.app.

- **More details:** emirtutar.github.io/Portfolio/#/projects/outdoor-planner
- **Technologies:** Vue.js, JavaScript, REST API, html, CSS, Git

Mission Fried Chicken (Team project)

Arcade game built to practise clean game architecture (game logic, UI, scene/menu structure, simple animations) with a focus on responsive controls and timing; implements the full game loop.

- **More details:** <https://emirtutar.github.io/Portfolio/#/projects/mission-fried-chicken>
- **Technologies:** C#, Azure DevOps (CI/CD build & test pipeline), Git

EDUCATION

B.Sc. Applied Computer Science (09/2021 – 03/2026)

Ravensburg Weingarten University, Grade: 2.3

Specialization: Robotics and Smart Devices

Fachabitur (09/2019 – 08/2021)

Fachoberschule, Lindau (Bavaria), Grade: 2.4

Six-month school-based internship in metalworking, woodworking, and electrical work

Mittlere Reife (09/2013 – 08/2019)

Realschule im Dreiländereck, Lindau (Bavaria), Grade: 2.5

Elected student representative in years 9 and 10

Notes

Availability: Immediately and open to relocation.